

PPE Detection with YOLOv8: Mini Report

Master's programme · Module 4 Unit 3: Computer Vision for AECO · 25 September 2026 · Repository:
github.com/argmanu/M4U3_ppe-detection-yolov8

Executive summary. I trained a YOLOv8n detector on 134 photos from my own workplace. It pre-screens images for valid **head protection** and **high-visibility clothing**. On the 20 % validation split, iteration 1 reaches **mAP50 0.923, precision 0.833 and recall 0.897**, which meets the success criteria set before evaluation (recall ≥ 0.85 per class, mAP50 ≥ 0.80). The model works well for single, close workers. It fails on small, distant and crowded scenes, and it did **not** learn the project rule that an open vest is not valid PPE.

Following the course feedback, iteration 2 adds explicit **violation classes**. It can flag a worker without a vest, which iteration 1 cannot do by design. Violations are rare in a compliant workplace, however, so iteration 2 remains a proof of concept. Everything runs in Google Colab without credentials: the dataset and weights are frozen as GitHub Release assets with SHA-256 checks.

1. PROBLEM AND APPROACH

Safety staff check PPE in site photos by eye, which is slow and inconsistent. The goal is an assistive screening step that highlights valid PPE, so a human reviewer can concentrate on doubtful cases. The two classes are defined in a written label contract. One deliberate academic rule is that an **open vest is not valid**: the model must separate wearing a garment from wearing it correctly.

2. DATA AND TRAINING

- 134 photos taken by the author. The people shown gave written consent. Faces were obscured in most images.
- Labels: Roboflow Astra Auto Label proposals, then a full manual review. 308 objects (177 head, 131 clothing).
- Split: 107 train / 27 validation (80/20). Preprocessing: auto-orient, fit within 640 px. No augmentation.
- 5 new test photos never uploaded to Roboflow. One candidate was rejected after I found it in the validation split.
- Model: YOLOv8n fine-tuned from COCO `yolo8n.pt` ; 30 epochs, imgsz 640, batch 16, seed 0, Ultralytics 8.2.103.
- Baseline:** the generic COCO model finds *person* but has no PPE classes, so a custom model is required.
- Reproducibility: Roboflow v3 export → Release v1.0 (SHA-256). A fresh-runtime Run all on Colab CPU took 1.6 min and gave mAP50 0.922.

3. RESULTS: ITERATION 1 (VALIDATION, 27 IMAGES, 70 OBJECTS)

CLASS	PRECISION	RECALL	MAP50	MAP50-95
all	0.833	0.897	0.923	0.717
head_protection	0.920	0.860	0.898	0.646
high_visibility_clothing	0.746	0.933	0.947	0.789

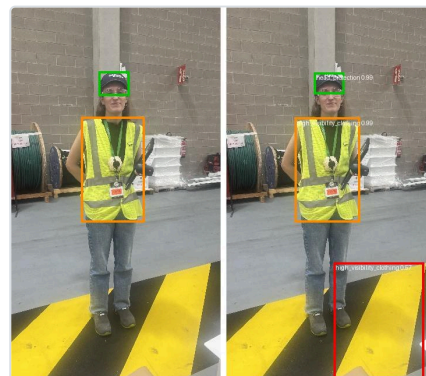
Interpretation. Clothing is found reliably, but it is over-detected: yellow and striped patterns trigger it. Caps are precise, but they are missed when small, distant or seen from above. With only 27 validation images, a single error moves recall by several points, so these metrics are indicative, not proof of production readiness.

4. ERROR ANALYSIS

- False positives:** yellow and black floor hazard stripes (0.57); a plain red T-shirt (0.35); a shadow next to a cap (0.37).
- False negatives:** distant caps in the background; several heads in a crowded scene; a cap seen from above.
- Open-vest rule not learned:** an open visitor vest (IMG_3474) and my open-vest test photo (0.52) were both accepted as valid.
- Label quality:** one open vest had been labelled as valid by mistake (IMG_3471).

Prioritised data improvements:

- Fix the label contract and relabel (done in iteration 2).
- Add small, distant and crowded workers.
- Add hard negatives: floor markings, red shirts, shadows.



False positive. Left: labels; right: prediction. The floor stripes are boxed as a vest (red).

5. ITERATION 2: FROM PPE DETECTION TO COMPLIANCE DETECTION

"No helmet detected" does not prove "no helmet worn". As the course feedback recommended, the violation became a class: `no_head_protection` (head box) and `no_high_visibility_clothing` (torso box, **including open vests**). The iteration uses the same images, split and recipe (Roboflow v5, Release v2.0). The new labels were drawn manually after a SAM 3 test (see section 6). A consistency check (one head label and one torso label per person) caught 6 open garments with missing or wrong labels before training. The model was trained on a Colab CPU in 30.1 min.

VALIDATION	ITERATION 1	ITERATION 2
mAP50 / P / R (all)	0.923 / 0.833 / 0.897	0.905 / 0.809 / 0.782
<code>head_protection</code> recall	0.860	0.713
<code>high_visibility_clothing</code> recall	0.933	0.897
<code>no_high_visibility_clothing</code> P / R	n/a	0.711 / 0.875
<code>no_head_protection</code> P / R	n/a	1.000 / 0.645*

*Only 3 validation examples (23 in total), so this is not statistically meaningful.

Findings. On a new photo of a forklift driver in a red T-shirt, iteration 2 **flags the missing vest**, which iteration 1 cannot express. On my open-vest photo it no longer accepts the vest, but it does not flag the violation either. The shared classes lost recall, and crowded scenes got worse. Violations are rare in a compliant workplace, which is the class imbalance the course feedback anticipated. **Conclusion:** iteration 1 remains the better PPE detector, and iteration 2 is a proof of concept. Next step: stage 30–50 photos of deliberate violations (bare heads, open vests) to balance the classes.



Same new photo. Left: iteration 1 (cap only). Right: iteration 2 flags `no_high_visibility_clothing` (0.66).

6. SAM 3 EXPLORATION

I tested SAM 3 in Roboflow (*Find Objects with AI*) with the prompt "upper body of a worker wearing a red t-shirt, without a reflective safety vest". **What helped:** precise masks from text alone, with no clicks or training. **What failed:** it also selected the worker who was wearing a vest, because absence cannot be prompted. The shoes came back as separate fragments, which would stretch the box (the mask-to-box "geometry tax"). It also cannot know project rules such as the open-vest rule. **Decision:** I did not use its suggestions. Rule-based and absence-based classes need human labels written against a contract.

7. LIMITATIONS, RISK AND GOVERNANCE

- **Scope:** one site, 134 images and 27 validation images. Not validated for crowded or distant scenes, other sites, weather or other PPE types (boots, glasses, harnesses).
- **Risk:** false negatives (missed PPE, or a missed violation) are the costly error, so thresholds should favour recall and a human must review every image. False positives cost reviewer time, and in a compliance setting they can hide a violation.
- **Do not use** for automated enforcement, discipline, surveillance or identification. With iteration 1, the absence of a detection does not prove the absence of PPE.
- **Privacy:** the photos are the author's own and written consent covers publication. Most faces are obscured. The model detects objects, not people. No identity or biometric data is processed.
- **Licensing:** code MIT; dataset CC BY 4.0 (own photos); Ultralytics YOLOv8 is AGPL-3.0.

8. REPRODUCIBILITY

Open either notebook from the README's *Open in Colab* badge, then choose *Runtime* → *Disconnect and delete runtime* → *Run all*. With a GPU the notebook retrains; without one it downloads the published `best.pt` and verifies its SHA-256. Dataset versions, model, epochs, batch, image size, seed and the Ultralytics version are listed in the README checklist, together with the reproducibility proof of each run.

This model is an assistive tool for preliminary screening only. It produces false negatives and false positives. It must not be used as the sole verifier for life-safety decisions.